MA1522 Linear Algebra for Computing

Lesson 7: Euclidean Spaces, Dot Product, and Linear Spans

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson7` (e.g. `A1234567Z_lesson7.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 7**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



The screen does not work, so I can't show the pollev page.

You can still ask question with this QR code.

Basic rules:

- Please (mainly) ask the questions related to the course.

Section 3.1 Euclidean Vector Spaces

The **Euclidean n-space** $\mathbb{R}^n$ is the collection of all n-vectors $\mathbf{v} = (v_i)$, where $v_i \in \mathbb{R}$ is the i-th **coordinate**. A vector is drawn as an arrow with its tail at the origin, or as the point at its head.

A set V with an addition and a scalar multiplication is a **vector space** over $\mathbb{R}$ if it satisfies these ten axioms. Here $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are in V and $a, b \in \mathbb{R}$.

1. $\mathbf{u} + \mathbf{v}$ is in V;
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$;
3. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$;
4. there is $\mathbf{0}$ in V with $\mathbf{0} + \mathbf{v} = \mathbf{v}$;
5. each $\mathbf{u}$ has $-\mathbf{u}$ in V with $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$;
6. $a\mathbf{v}$ is in V;
7. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$;
8. $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$;
9. $a(b\mathbf{u}) = (ab)\mathbf{u}$;
10. $1\mathbf{u} = \mathbf{u}$.

Since vectors are matrices, matrix algebra gives all ten for $\mathbb{R}^n$.

Section 3.2 Dot Product, Norm, Distance

The **dot product** (or **inner product**) of $\mathbf{u} = (u_i)$ and $\mathbf{v} = (v_i)$ in $\mathbb{R}^n$ is

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

Norm, unit vector

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{u_1^2 + \cdots + u_n^2}.$$

$\mathbf{u}$ is a **unit vector** if $\|\mathbf{u}\| = 1$; for $\mathbf{u} \neq \mathbf{0}$, **normalizing** gives one: $\mathbf{u}/\|\mathbf{u}\|$.

Distance, angle

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

For $\mathbf{u}, \mathbf{v} \neq \mathbf{0}$, the **angle** θ between them, with $0 \leq \theta \leq \pi$, is given by $\cos\theta = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}$.

The dot product is symmetric and distributes over sums, and $\mathbf{u} \cdot \mathbf{u} \geq 0$ with equality only for $\mathbf{u} = \mathbf{0}$. Scalars come out of both: $\mathbf{u} \cdot (c\mathbf{v}) = c\,(\mathbf{u} \cdot \mathbf{v})$ and $\|c\mathbf{u}\| = |c| \, \|\mathbf{u}\|$.

Section 3.3 Linear Combinations and Linear Spans

Convention: Span(empty set) := {0}

Let $\mathbf{u}_1, \ldots, \mathbf{u}_k$ be vectors in $\mathbb{R}^n$. A **linear combination** of them is $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_k\mathbf{u}_k$, where the scalars $c_1, \ldots, c_k \in \mathbb{R}$ are the **coefficients**. Their **span** is the set of all linear combinations,

$$\operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} = \{\, c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_k\mathbf{u}_k \mid c_1, \ldots, c_k \in \mathbb{R} \,\} \subseteq \mathbb{R}^n.$$

$\mathbf{w} \in$ ... This means that $\exists\, c_1, \ldots, c_k$ such that $\mathbf{w} = c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k$.

In other words $\begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k \\ | & | & & | \end{pmatrix} \begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix} = \mathbf{w}$, i.e. A $\quad$ $\mathbf{x}$

Algorithm: is $\mathbf{v}$ a linear combination?

Form the $n \times k$ matrix $\begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_k \end{pmatrix}$ with the vectors as its columns. Then

$$\mathbf{v} \in \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \iff \left(\begin{array}{ccc|c} \mathbf{u}_1 & \cdots & \mathbf{u}_k & \mathbf{v} \end{array} \right) \text{ is consistent,}$$

and the solutions of that system are exactly the possible coefficients.

Question 1

1: Whether we can write.

$v = c_1 w_1 + c_2 w_2 + c_3 w_3$ for some $c_1, c_2, c_3 \in \mathbb{R}$.

I.e. whether. $\begin{pmatrix} | & | & | \\ w_1 & w_2 & w_3 \\ | & | & | \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = v$ is consistent?

Let

$$\mathbf{w}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{w}_3 = \begin{pmatrix} 1 \\ -1 \\ -3 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}.$$

1. Is $\mathbf{v}$ in $\text{span}\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$? If it is, write $\mathbf{v}$ as a linear combination of $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$.

2. Are the coefficients unique? If not, give two different sets of coefficients.

⇑

Does the system $\left(\begin{array}{ccc|c} | & | & | & \\ w_1 & w_2 & w_3 & X = v \\ | & | & | & \end{array}\right)$ have a unique sol.?

Question 1 — solution: part (1)

1: Whether we can write.

$$v = c_1 w_1 + c_2 w_2 + c_3 w_3 \quad \text{for some } c_1, c_2, c_3 \in \mathbb{R}.$$

I.e. whether $\begin{pmatrix} | & | & | \\ w_1 & w_2 & w_3 \\ | & | & | \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = v$ is consistent?

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 1 & -1 & 5 \\ 3 & 0 & 3 & 6 \\ \| & \| & \| & \| \\ w_1 & w_2 & w_3 & v \end{array}\right) \xrightarrow{\text{ref.}} \left(\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

consistent. ($\Leftrightarrow v \in \mathrm{Span}\{w_1, w_2, w_3\}$)

Find a solution: $\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 2+t \\ 1-t \\ t \end{pmatrix}$ $\forall\, t \in \mathbb{R}$.

Take $t=0$. $\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$.

Check: $c_1 w_1 + c_2 w_2 + c_3 w_3 \overset{?}{=} v$.

$$2\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + 1\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + 0\cdot\begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = v.$$

Question 1 — solution: part (2)

To get another solution, we can take t to be any other real numbers. E.g.

$t = 1$, $\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 2+t \\ 1-t \\ t \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$ $\square$

Section 3.3 When is span$(S) = \mathbb{R}^n$?

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\} \subseteq \mathbb{R}^n$. View the vectors as column vectors, and let $A = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k \end{pmatrix}$. Then

$$
\begin{aligned}
\operatorname{span}(S) = \mathbb{R}^n &\iff \text{for all } \mathbf{v} \in \mathbb{R}^n,\ A\mathbf{x} = \mathbf{v} \text{ has a solution} \\
&\iff \text{for all } \mathbf{v} \in \mathbb{R}^n,\ A\mathbf{x} = \mathbf{v} \text{ is consistent} \\
&\iff \operatorname{RREF}(A) \text{ has no zero row} \\
&\iff \text{all rows of } \operatorname{RREF}(A) \text{ are nonzero} \\
&\iff \text{all rows of any row-echelon form of } A \text{ are nonzero.}
\end{aligned}
$$

Example 1

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ be a set of k vectors in $\mathbb{R}^n$. True or false?

1. If $k < n$, then $\text{span}(S) \neq \mathbb{R}^n$.
2. If $k \geq n$, then $\text{span}(S) = \mathbb{R}^n$.

Explain your answer to your neighbour.

2. Counter example.

$S = \{\begin{pmatrix}0\\0\\0\end{pmatrix}, \begin{pmatrix}0\\0\\0\end{pmatrix}, \ldots, \begin{pmatrix}0\\0\\0\end{pmatrix}\}$, $\text{span}(S) = \{\begin{pmatrix}0\\0\\0\end{pmatrix}\}$ in $\mathbb{R}^3$.

k copies

1. $A = \begin{pmatrix} | & & | \\ u_1 & \cdots & u_k \\ | & & | \end{pmatrix}$ $k \times k$ n $\xrightarrow{\text{ref}}$ $k \times k$ n with a zero row k

so $\text{rref}(A)$ must have a zero row

$\Rightarrow \text{span}(S) \subsetneq \mathbb{R}^n$

Question 2

Let

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 3 \\ 6 \\ -3 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix}.$$

1. Show that $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \mathbb{R}^3$, and write each of $\mathbf{v}$ and $\mathbf{w}$ as a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$.

2. For $i = 1, 2, 3$, compute $\dfrac{\mathbf{v} \cdot \mathbf{u}_i}{\|\mathbf{u}_i\|^2}$ and $\dfrac{\mathbf{w} \cdot \mathbf{u}_i}{\|\mathbf{u}_i\|^2}$, and compare them with your coefficients from part (1). What do you notice?

Question 2 — solution: part (1)

$$\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix} \xrightarrow{\text{ref}} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \Longrightarrow \operatorname{span}(\{u_1, u_2, u_3\}) = \mathbb{R}^3 .$$

u_1 u_2 u_3

$c_1 = 1,\ c_2 = 2,\ c_3 = -1$

$c_1 = 2,\ c_2 = -1,\ c_3 = 1.$

$$\left(\begin{array}{ccc|c|c} 1 & 2 & 2 & 3 & 2 \\ 2 & 1 & -2 & 6 & 1 \\ 2 & -2 & 1 & -3 & 7 \end{array}\right) \xrightarrow{\text{ref}} \left(\begin{array}{ccc|c|c} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 1 \end{array}\right)$$

u_1 u_2 u_3 v w

This means that.

$$v = \begin{pmatrix} 3 \\ 6 \\ -3 \end{pmatrix} = 1 \cdot u_1 + 2 \cdot u_2 + (-1) \cdot u_3 .$$

$$w = \begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix} = 2 \cdot u_1 + (-1) \cdot u_2 + 1 \cdot u_3 .$$

Question 2 — solution: part (2)

Compute $\dfrac{v \cdot u_i}{\|u_i\|^2}$.

$v = \begin{pmatrix} 3 \\ 6 \\ -3 \end{pmatrix}$. $u_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$. $u_2 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$. $u_3 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$.

- $v \cdot u_1 = 3 \cdot 1 + 6 \cdot 2 + (-3) \cdot 2 = 9$.
- $\|u_1\|^2 = \sqrt{u_1 \cdot u_1}^2 = u_1 \cdot u_1 = 1 \cdot 1 + 2 \cdot 2 + 2 \cdot 2 = 9$.

$\dfrac{v \cdot u_1}{\|u_1\|^2} = \dfrac{9}{9} = 1$.

$\dfrac{v \cdot u_2}{\|u_2\|^2} = 2$. $\dfrac{v \cdot u_3}{\|u_3\|^2} = -1$.

Remark:

One can check that. $\{u_1, u_2, u_3\}$ is orthogonal. Ie. $u_i \cdot u_j = 0$ if $i \neq j$.

In this situation, we can compute c_1, c_2, c_3 through dot product directly.

Question 3

Let

$$S_1 = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}, \qquad S_2 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \right\}.$$

1. Determine whether each of S_1 and S_2 spans $\mathbb{R}^3$.

2. For the set that does not, find a vector of $\mathbb{R}^3$ that is not in its span.

Question 3 — solution: part (1)

(1). $\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \Rightarrow \operatorname{span}(S_1) = \mathbb{R}^3$.

$\begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 2 & 1 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \operatorname{span}(S_2) \subsetneq \mathbb{R}^3$.

(2) We want to find a vector $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ s.t.

$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & x \\ 0 & 1 & 1 & -1 & y \\ 1 & 1 & 2 & 1 & z \end{array}\right)$ ~ row operations ~ $\left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & * \\ 0 & 1 & 1 & -1 & * \\ 0 & 0 & 0 & 0 & * \end{array}\right)$

↳ is nonzero.

Question 3 — solution: part (2)

Method 1: Do row operations by hand.

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & x \\ 0 & 1 & 1 & -1 & y \\ 1 & 1 & 2 & 1 & z \end{array}\right) \xrightarrow{R_3 - R_1} \left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & x \\ 0 & 1 & 1 & -1 & y \\ 0 & 1 & 1 & -1 & z-x \end{array}\right) \xrightarrow{R_3 - R_2} \left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & x \\ 0 & 1 & 1 & -1 & y \\ 0 & 0 & 0 & 0 & z-x-y \end{array}\right)$$

This system is inconsistent $\iff z - x - y \neq 0$.

In other words, $\operatorname{span}(S_2) = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \;\middle|\; z - x - y = 0 \right\}$

Method 2: Challenge: interpret this.

$$\left(\begin{array}{cccc|ccc} 1 & 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{\text{rref}} \left(\begin{array}{cccc|ccc} 1 & 0 & 1 & 2 & 0 & -1 & 1 \\ 0 & 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & -1 \end{array}\right) \rightsquigarrow 1 \cdot x + 1 \cdot y + (-1) z$$

Next lesson

- **Sections 3.3 & 3.4 Linear Spans and Subspaces**

Please watch the video clips for these sections beforehand.